# Computer Networks Lab

## [ENCS352]

## KR. MANGALAM UNIVERSITY

**For**

## BACHELORS IN TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

**With Specialization In AI & ML**

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| S. No. | Experiment Title | Date | Remarks / Sign |
|---|---|---|---|
| 1 | (a) CPT Platform Overview<br>(b) Peer to Peer Network | 22/01/26 | |
| 2 | Network Topologies – (a) Star Topology | 29/01/26 | |
| | Network Topologies – (b) Ring Topology | 29/01/26 | |
| | Network Topologies – (c) Mesh Topology | 05/02/26 | |
| | Network Topologies – (d) Tree Topology | 05/02/26 | |
| 3 | (a) LAN using Hub | 19/02/26 | |
| | (b) LAN using Switch | 19/02/26 | |
| 4 | Tracert and Ping commands – Network Diagnostics | 26/02/26 | |
| 5 | (a) Designing the Network (CAN) | 19/03/26 | |
| | (b) Static IP Address Assignment to devices in the Network | 02/04/26 | |
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# EXPERIMENT 1

## (a): CPT PLATFORM OVERVIEW

### Theory

Cisco Packet Tracer is a network simulation tool used to design and test network topologies without physical hardware. It provides a virtual environment with devices like routers, switches, and PCs. It helps students understand networking concepts. It supports both real-time and simulation modes. It is widely used for learning and experimentation.

### Steps

1. Open Cisco Packet Tracer.
2. Observe workspace, device toolbar, and connections panel.
3. Drag one PC and one switch.
4. Connect them using a straight-through cable.
5. Click on devices to explore Desktop and Config tabs.

### Result

Successfully explored the Cisco Packet Tracer interface.

### Conclusion

Packet Tracer provides an effective platform for learning networking concepts.

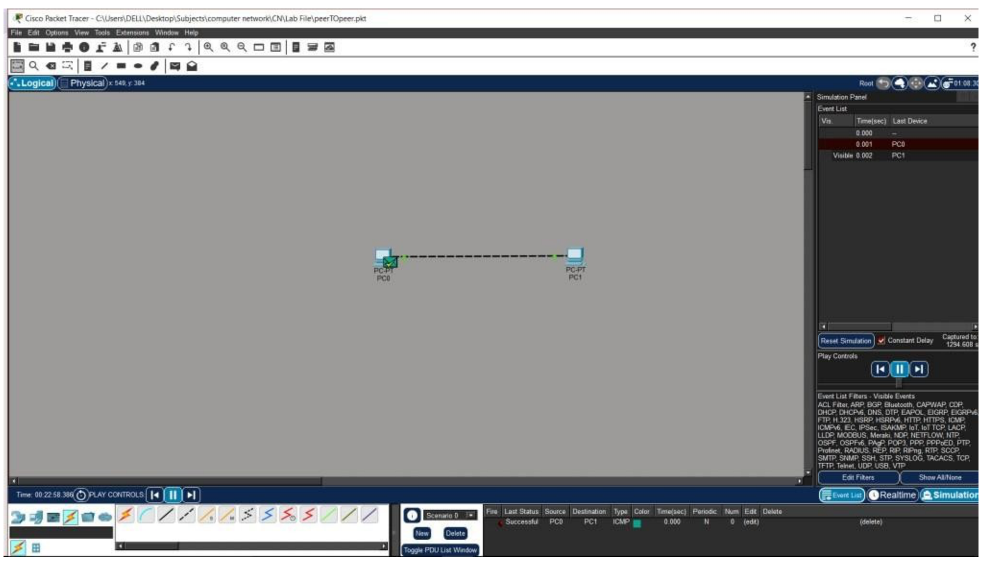
# (b): PEER-TO-PEER NETWORK

## Theory

A peer-to-peer network allows devices to communicate directly without a central server. Each device acts as both client and server. It is simple, cost-effective, and suitable for small networks. Data sharing can be done directly between devices. However, it lacks centralized control and security.

## Steps

1. Add two PCs (PC0, PC1).
2. Connect using a cross-over cable.
3. Assign IP addresses:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
4. Open Command Prompt on PC0.
5. Run: ping 192.168.1.2



## Result

Ping successful between PCs. Peer-to-peer communication was established successfully.

# EXPERIMENT 2

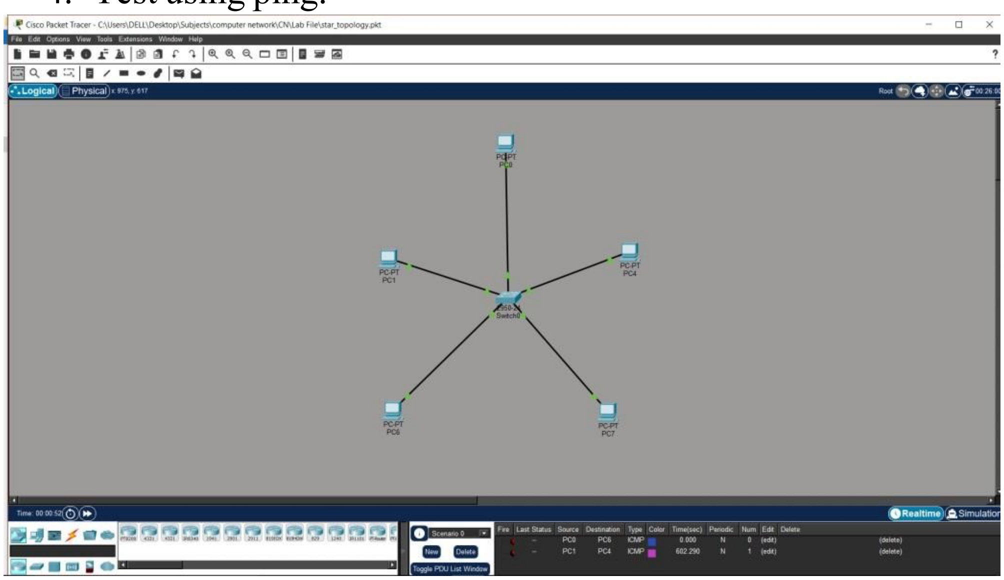
## (a): STAR TOPOLOGY

### Theory

Star topology connects all devices to a central switch. All communication passes through this central device. It is easy to install and manage. Failure of one link does not affect others. However, failure of the central device stops the entire network.

### Steps

1. Add one switch and multiple PCs.
2. Connect PCs to switch using straight cable.
3. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
    d. PC3 → 192.168.1.4
4. Test using ping.




### Result

All devices communicate through switch. Star topology is reliable and easy to troubleshoot.

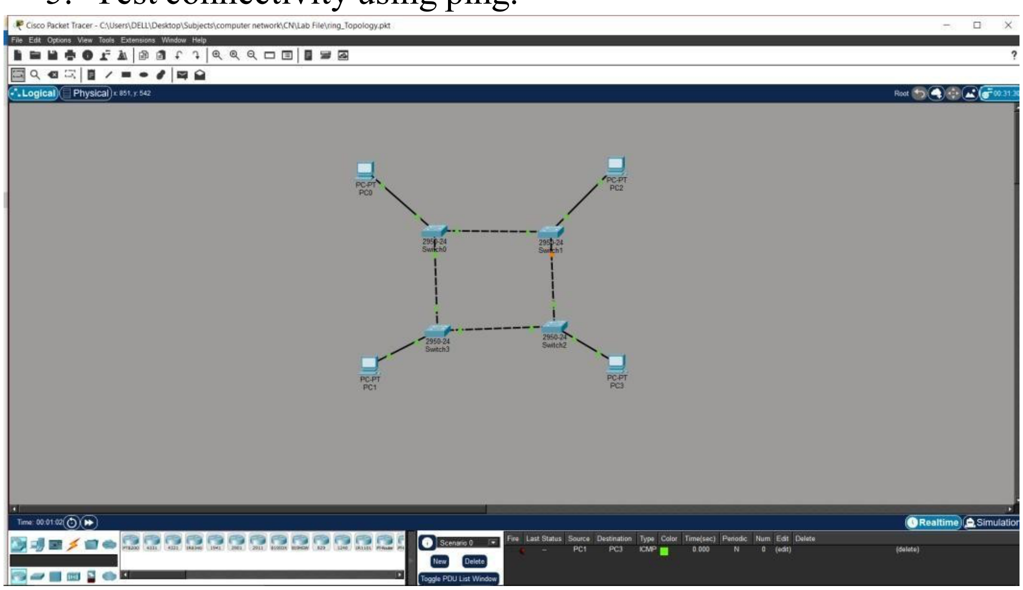
# (b): RING TOPOLOGY

## Theory

Ring topology connects devices in a circular form where each node is connected to two others. Data travels in one direction around the ring. It ensures equal access to all nodes. However, failure of a single link can disrupt the entire network. It is less commonly used in modern networks.

## Steps

1. Add 4 PCs and 4 switches.
2. Connect switches in circular form.
3. Connect PCs to switches.
4. Assign IPs:
   a. PC0 → 192.168.1.1
   b. PC1 → 192.168.1.2
   c. PC2 → 192.168.1.3
   d. PC3 → 192.168.1.4
5. Test connectivity using ping.




## Result

Communication is successful in ring structure. Ring topology works but is less reliable than star.

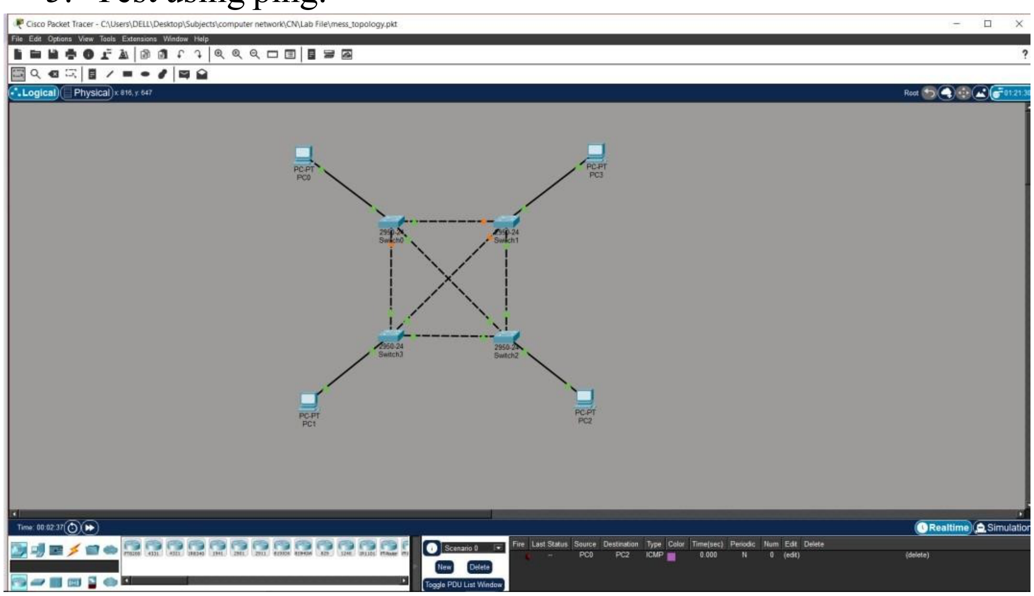
# (c): MESH TOPOLOGY

## Theory

Mesh topology connects every device to every other device. It provides multiple paths for data transmission. This increases reliability and fault tolerance. Even if one path fails, communication continues. However, it is expensive and complex to implement.

## Steps

1. Add switches and PCs.
2. Connect each switch with every other switch.
3. Connect PCs to switches.
4. Assign IPs:
   a. PC0 → 192.168.1.1
   b. PC1 → 192.168.1.2
   c. PC2 → 192.168.1.3
   d. PC3 → 192.168.1.4
5. Test using ping.



## Result

Multiple communication paths are established. Mesh topology provides high reliability.

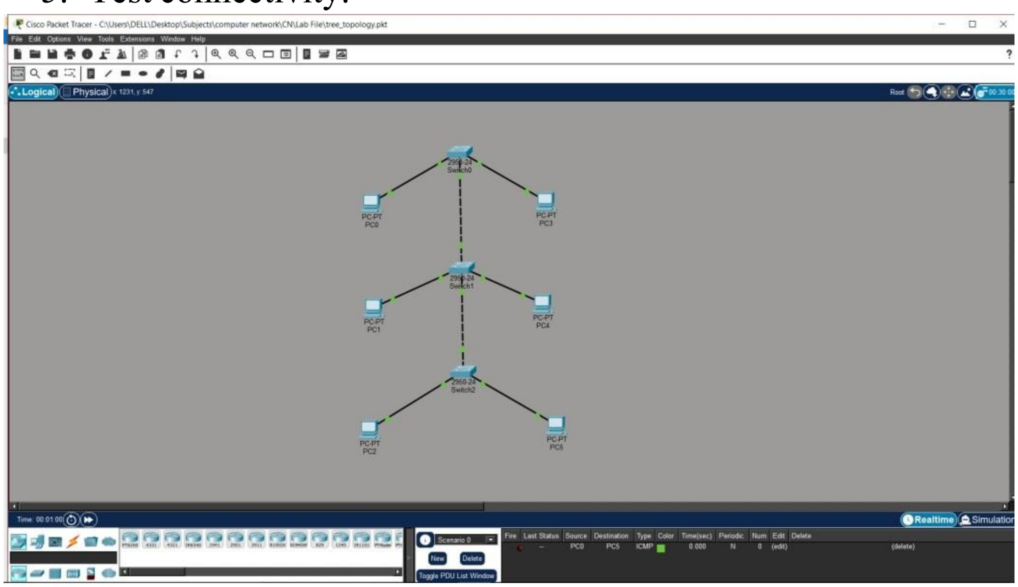
# (d): TREE TOPOLOGY

## Theory

Tree topology is a hierarchical network structure combining multiple star topologies. It has a root node and multiple levels. It is suitable for large networks. It allows for easy expansion and management. However, failure of backbone can affect the entire network.

## Steps

1. Add multiple switches in hierarchy.
2. Connect switches in parent-child structure.
3. Connect PCs to switches.
4. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
    d. PC3 → 192.168.1.4
    e. PC4 → 192.168.1.5
    f. PC5 → 192.168.1.6
5. Test connectivity.




## Result

Hierarchical communication was established. Tree topology is scalable and efficient.

# EXPERIMENT 3

## (a): LAN USING HUB

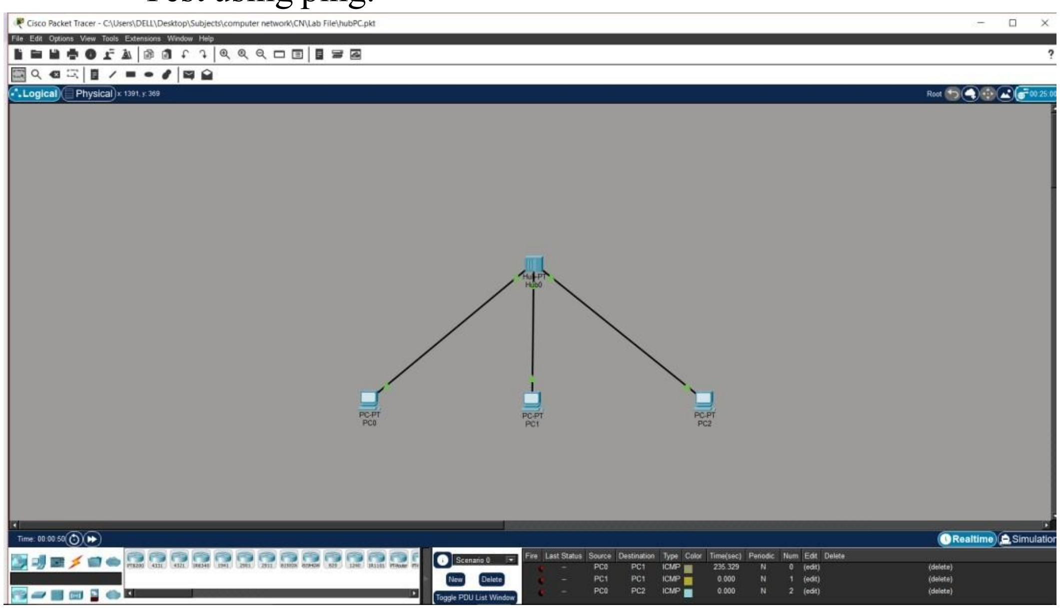
### Theory

A hub is a basic networking device that broadcasts data to all connected devices. It operates on the physical layer. It does not filter data, leading to collisions. It is simple but inefficient. It is mostly replaced by switches.

### Steps

1. Add one hub and three PCs.
2. Connect PCs to hub.
3. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3 4.

Test using ping.



### Result

All devices receive transmitted data. Hub-based LAN is inefficient due to broadcasting.

# (b): LAN USING SWITCH

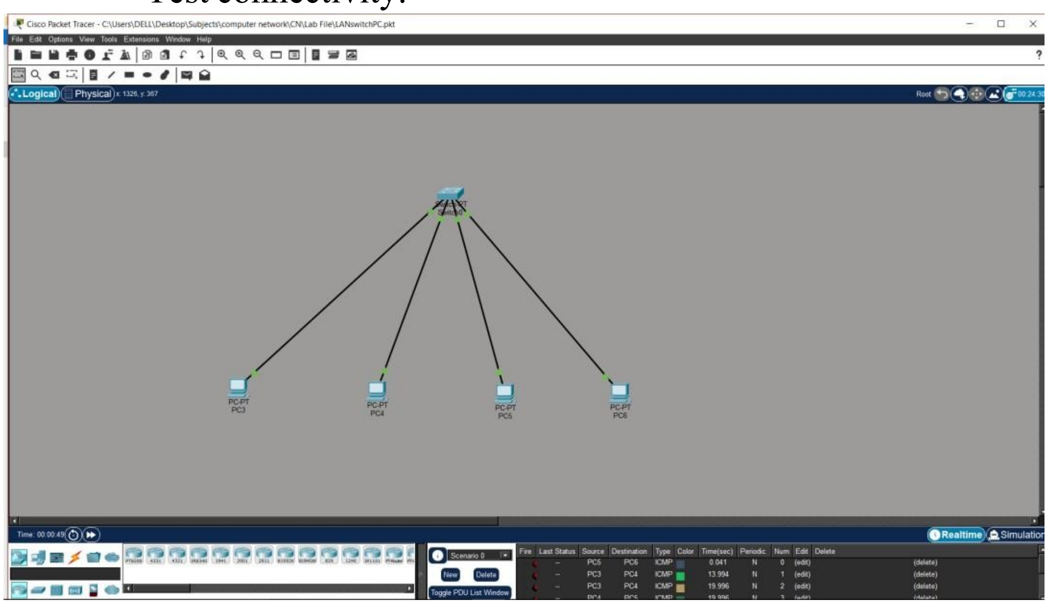
## Theory

A switch is a networking device that forwards data only to the destination device. It operates on the data link layer. It reduces collisions and improves performance. It uses MAC address tables for efficient communication. It is widely used in LANs.

## Steps

1. Add one switch and multiple PCs.
2. Connect PCs to switch.
3. Assign IPs:
    a. PC3 → 192.168.1.1
    b. PC4 → 192.168.1.2
    c. PC5 → 192.168.1.3
    d. PC6 → 192.168.1.4 4.

Test connectivity.




## Result

Efficient communication is achieved. Switches improve network performance.

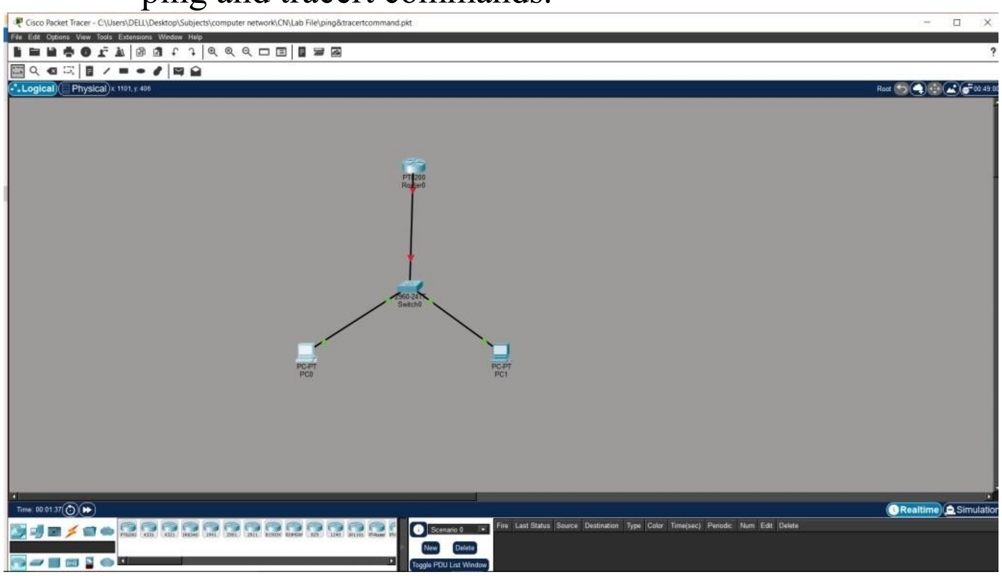
# EXPERIMENT 4

## PING & TRACERT

### Theory

Ping is used to test connectivity between devices using the ICMP protocol. Tracert shows the path taken by packets. These tools help troubleshoot network issues. They provide information about delays and failures. They are essential diagnostic tools. **Steps**

1. Add router, switch, and PCs.
2. Connect devices properly.
3. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. Router → 192.168.1.254 4. Set gateway in PCs → 192.168.1.254 5. Run ping and tracert commands.



### Result

Connectivity and path verified successfully. Ping and tracert help diagnose network problems.

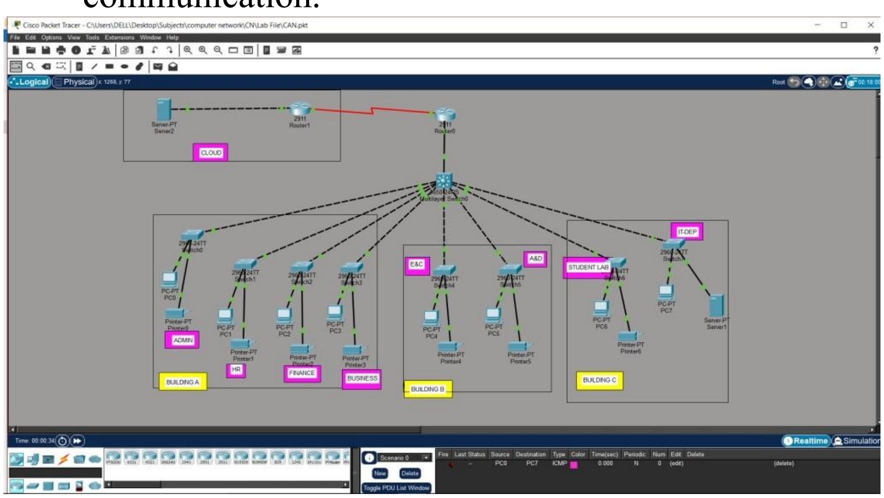
# EXPERIMENT 5

## CAMPUS AREA NETWORK (CAN)

### Theory

A Campus Area Network connects multiple LANs across buildings. It includes routers, switches, and servers. It allows communication between departments. **Steps**

1. Add router, multilayer switch, switches, PCs, servers.
2. Divide network into departments.
3. Connect all switches to central switch.
4. Assign IPs:
    a. ADMIN → 192.168.1.1
    b. HR → 192.168.2.1
    c. FINANCE →
192.168.3.1
    d. BUSINESS →
192.168.4.1
    e. E&C → 192.168.5.1
    f. A&D → 192.168.6.1
    g. STUDENT LAB →
192.168.7.1 5. Set gateways →
192.168.X.254 6. Test
communication.




**Result -** All departments connected successfully. CAN enables efficient campus-wide communication.

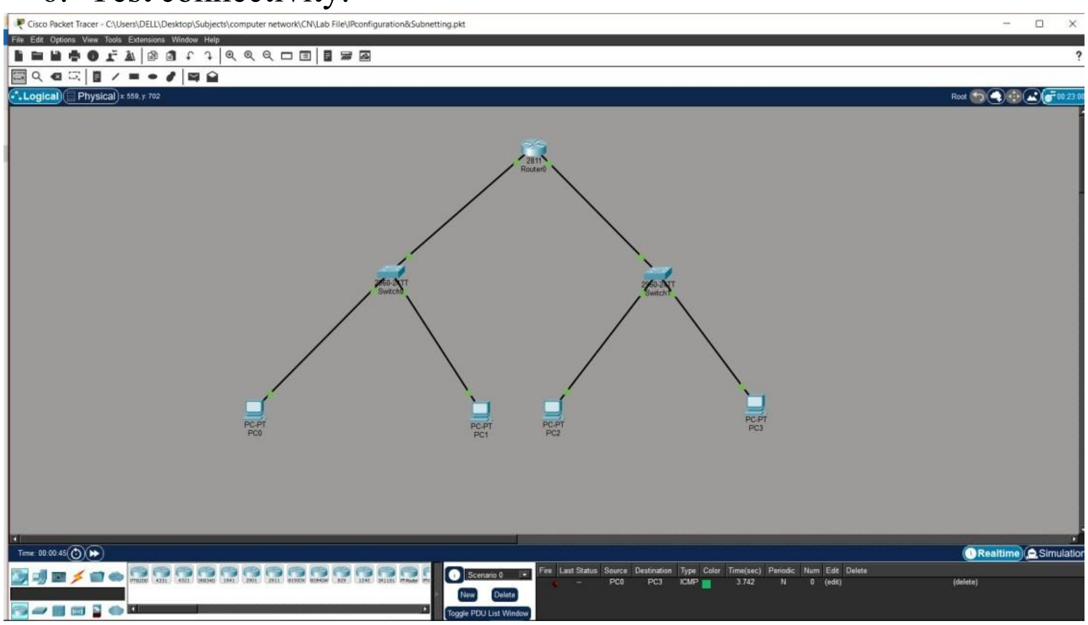
# EXPERIMENT 6

## IP CONFIGURATION & SUBNETTING

### Theory

IP addressing uniquely identifies devices in a network. Subnetting divides a large network into smaller subnetworks. It improves performance and security. Each subnet has its own range. It reduces congestion and improves management. **Steps**

1. Add router, switches, and PCs.
2. Divide network:
    a. Subnet1 → 192.168.1.0/25
    b. Subnet2 → 192.168.1.128/25
3. Assign IPs accordingly.
4. Configure router interfaces.
5. Set the gateways.
6. Test connectivity.




### Result

Subnetted network works correctly. Subnetting improves efficiency and organization.

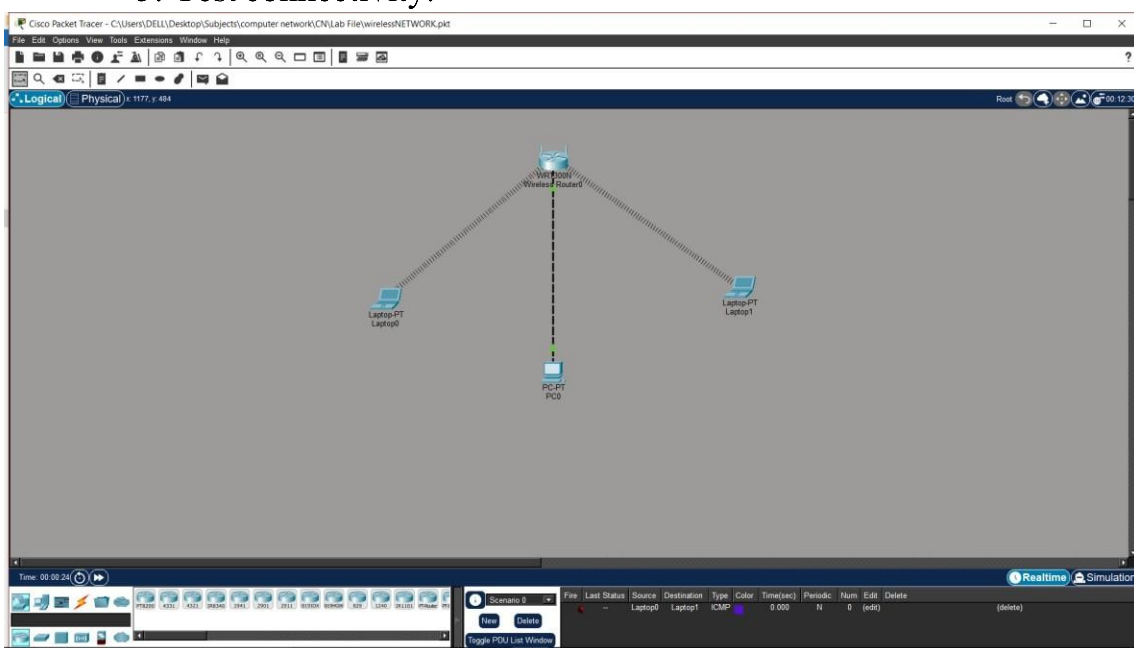
# EXPERIMENT 7

## WIRELESS NETWORK

### Theory

Wireless networks connect devices without cables using radio signals. A wireless router provides connectivity through SSID. Devices like laptops connect using Wi-Fi. It provides flexibility and mobility. It is widely used in homes and offices. **Steps**

1. Add wireless router and laptops.
2. Configure SSID and password.
3. Connect laptops wirelessly.
4. Assign IPs:
    a. Router → 192.168.0.1
    b. Laptop0 → 192.168.0.2
    c. Laptop1 → 192.168.0.3
5. Test connectivity.




### Result

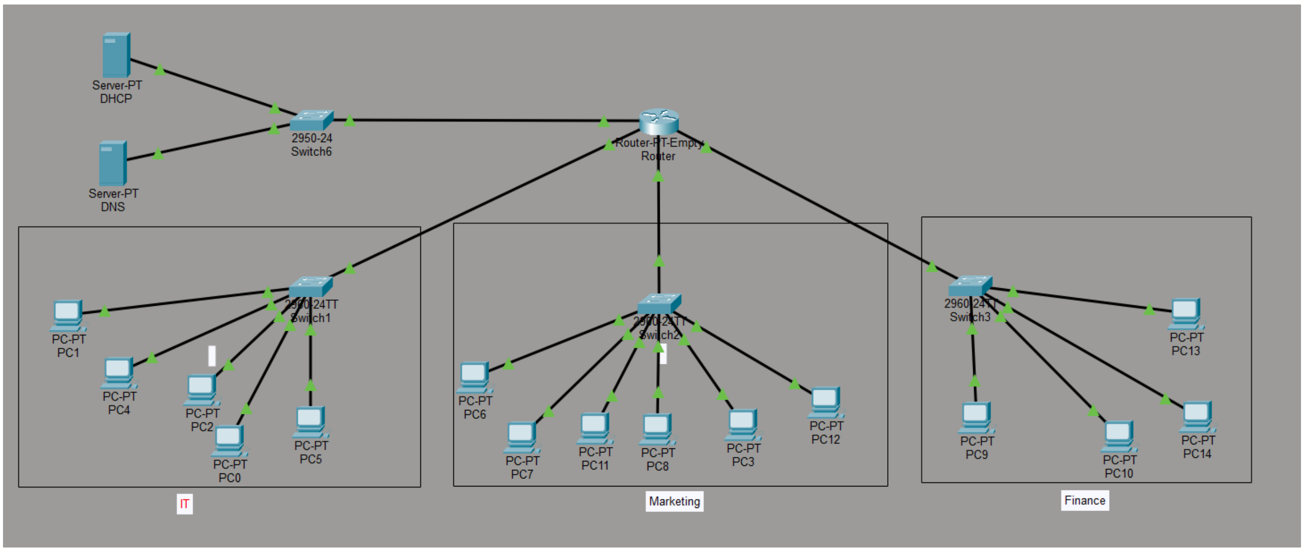
Wireless communication is established. Wireless networks provide flexibility and ease of use.

# EXPERIMENT 8

## CAPSTONE PROJECT

## Objective:

- The objective of this network is to design and implement a **small enterprise LAN** that:
- Connects multiple departments (**IT, Marketing, Finance**)
- Provides **centralized IP management using DHCP**
- Enables **name resolution using DNS**
- Ensures **inter-department communication via a router**
- Maintains scalability and organized segmentation




## Requirements:

- **End Devices (15 PCs):** Configured to automatically request IP addresses.
  - IT Department: 5 PCs
  - Marketing Department: 6 PCs
  - Finance Department: 4 PCs
- **Servers (2):**
  - **DHCP Server:** Automates the distribution of IPs, subnet masks, gateways, and DNS records.

    - **DNS Server:** Resolves human-readable domain names into internal IP addresses.
- **Connecting Devices:**
    - **Router (1x Router-PT-Empty):** Acts as the core gateway routing traffic between the four subnets. Requires physical interface modules to be installed manually.
    - **Switches (4 Total):** Layer 2 devices establishing local broadcast domains. Uses one 2950-24 for the servers and three 2960-24TT switches for the departments.
- **Media:**
    - **Copper Straight-Through Cables:** Used for all connections (PC-to-Switch, Server-to-Switch, Switch-to-Router).

## Implementation Steps:

- **Hardware Deployment:** Place the router, switches, servers, and PCs onto the workspace.
- **Router Module Setup:** Power off the router, install four FastEthernet or GigabitEthernet modules into the empty slots, and power it back on.
- **Cabling:** Connect all devices using straight-through cables.
- **Gateway Configuration:** Assign specific IP addresses to the four router interfaces to act as Default Gateways for each subnet, and enable them (no shutdown).
- **DHCP Relay (Crucial):** Configure the ip helper-address [DHCP Server IP] command on the router interfaces connected to IT, Marketing, and Finance to forward broadcast DHCP requests to the server subnet.
- **Server Configuration:**
    - Assign static IP addresses to both servers.
    - Create three distinct address pools on the DHCP server for the departments.
    - Create necessary internal A-Records on the DNS server.
- **Client Configuration:** Set the IP Configuration on all 15 PCs from "Static" to "DHCP".
- **Verification:** Use the ping command from a PC to test connectivity across the different departmental subnets.

## IP Configuration Scheme:

- **Server Subnet:** 192.168.10.0/24
  - Gateway: 192.168.10.1 | DHCP: 192.168.10.2 (Static) | DNS: 192.168.10.3 (Static)
- **IT Subnet:** 192.168.20.0/24
  - Gateway: 192.168.20.1 | PC Range: 192.168.20.2 - 254 (DHCP)
- **Marketing Subnet:** 192.168.30.0/24
  - Gateway: 192.168.30.1 | PC Range: 192.168.30.2 - 254 (DHCP)
- **Finance Subnet:** 192.168.40.0/24
  - Gateway: 192.168.40.1 | PC Range: 192.168.40.2 - 254 (DHCP)

## Estimated Budget:

| **Item** | **Description** | **Quantity** | **Unit Cost (Est.)** | **Total Cost (Est.)** |
|---|---|---|---|---|
| **Core Router** | Refurbished Cisco 1941 or Mikrotik Cloud Core | 1 | ₹28,000 | **₹28,000** |
| **Network Switches** | Refurbished Cisco 2960 or New TP-Link Managed | 4 | ₹9,500 | **₹38,000** |
| **Department Servers** | Assembled Tower Servers (i5/16GB/1TB) | 2 | ₹45,000 | **₹90,000** |
| **Workstations (PCs)** | Assembled Office PCs (i3/8GB/SSD) | 15 | ₹26,000 | **₹3,90,000** |
| **Passive Networking** | Cat6 Cable Roll (305m), I/O Boxes, Racks | 1 | ₹22,000 | **₹22,000** |

**Total Estimated Cost**- ₹5,68,000

## Learning Outcomes:

- **Network Segmentation:** Breaking down a large network into manageable, isolated departmental broadcast domains.
- **Service Centralization:** Using dedicated servers for DHCP and DNS instead of decentralized router configurations.
- **Inter-Subnet Routing:** Directing traffic between physically and logically separated networks using a core router.
- **IP Helper Application:** Understanding how to pass broadcast traffic (like DHCP requests) across router interfaces.